

Feeling Understood Is Not Feeling Ownership:

Within- and Between-Writer Associations of Perceived Understanding, Ideation Help, and Ownership in the CoAuthor Dataset

Trial project, Option 2 — exploratory secondary analysis

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Abstract

Does feeling that an AI writing assistant understands one’s intent go together with feeling that the resulting text is one’s own? We linked the public CoAuthor metadata and post-session surveys and analyzed 1,407 sessions in which writers queried GPT-3 at least once (61 observed writer IDs). Perceived understanding and human-written share were decomposed into a writer mean and a within-writer deviation, and ownership was modeled with a cumulative-logit, working-independence GEE with prompt fixed effects, model-setting indicators, and writer-clustered sandwich errors. The hypothesized positive within-writer association between understanding and ownership was not supported (OR = 0.847, 95% CI [0.660, 1.086]). In contrast, a session with higher-than-usual understanding was strongly associated with higher reported ideation help (OR = 3.396 [2.761, 4.176]), and a 10-percentage-point higher-than-usual human-written share was associated with higher ownership (OR = 1.327 [1.130, 1.559]). The between-writer association of understanding with ownership was positive in the main model but attenuated under equal-writer weighting. Preliminary coding of 23 contrastively sampled comments showed useful support co-occurring with low ownership (“more an editor than a writer”) and intent mismatch co-occurring with high ownership. We argue that ideation support and ownership should be evaluated as separate outcomes, and propose *role alignment*—whether the writer’s desired and experienced roles match—as a follow-up question. All results are observational associations from single-item post-hoc ratings.

1 Introduction

What users value in co-creative writing may not be only the quality of the final text, but whether the text still carries *their* intent: which direction they chose, which suggestions they accepted, and how they revised them. If “good AI use” is defined by usage volume or output fluency alone, the experience of exploring with AI while remaining the author of one’s work is invisible.

Our starting motivation is respect for the individual’s own intent and values in human–AI collaboration, related to autonomy in self-determination theory [4]. CoAuthor [1] contains no validated measures of authenticity, personality, intrinsic motivation, or well-being, so we restrict the question to three observable post-session ratings: *the system understood what I was trying to write* (perceived understanding, *U*), *the suggestions helped me come up with new ideas* (ideation help, *I*), and *I feel like the story/essay is mine* (ownership, *O*). Prior work shows that AI-generated text can lower perceived ownership even when users still declare authorship [2], motivating us to treat ownership as distinct from both helpfulness and the share of text typed by the human.

Because the same writers completed many sessions (median 9, range 1–89), we separate two questions that are easily conflated [3]: **between writers**, do writers who generally feel understood report more ownership? and **within writers**, in sessions where the same person felt *more understood than usual*, did they also feel more ownership? The within-writer question matches our interest in situational variation of collaboration.

Hypotheses. Directions were fixed after inspecting the data structure but before the first model run (not preregistered).

- **H1 (primary).** Within-writer perceived understanding is positively associated with ownership, adjusting for human-written share, prompt, and model setting.
- **H2.** Within-writer perceived understanding is positively associated with ideation help.
- **H3.** The within-writer association between ideation help and ownership is more positive when within-writer understanding is higher ($U^W \times I^W$).

- **RQ4 (qualitative).** In sessions where understanding and ownership agree or diverge, how do writers describe support, mismatch, choice, and role?

H1 is the single primary test; H2–H3 are Holm-corrected as one family.

2 Data

We downloaded the four official sheets (creative/argumentative metadata and surveys) as XLSX on 2026-10-09 (SHA-256 `bb850549...47c4`). Table 1 gives the sample flow. Sessions were linked one-to-one by trimmed session ID within genre; worker IDs agreed for all linked sessions. Duplicate survey submissions kept the latest response; unmatched surveys were not imputed or fuzzy-matched. Sessions with zero queries (37) were treated as observed non-use, not missing data, and excluded from the main sample.

Table 1: Sample flow.

Step	n	Note
Metadata sessions	1,445	creative 830, argumentative 615
Survey rows	1,458	creative 833, argumentative 625
excess duplicate survey rows	−3	latest response kept
surveys with no metadata match	−11	not imputed
Linked sessions	1,444	1 metadata session had no survey
zero-query sessions	−37	observed non-use
Analysis sample	1,407	creative 823, argumentative 584; 61 writer IDs

Discrepancies to resolve. The paper reports 63 writers, whereas the public metadata contain 61 unique IDs; the paper describes ownership on a 5-point scale, whereas the downloaded responses range 1–7. We used the observed 1–7 coding without recoding and flag both for confirmation with the dataset authors.

Variables. U , I , O : 1–7 ratings. H : human-written share of the final text (0–100%, entered per 10 pp). Prompt (20 levels) enters as fixed effects; model setting is the genre-specific high/low decoding configuration (creative: $T=.30/FP=0$ vs. $T=.75/FP=1$; argumentative: $T=.20$ vs. $.90$, $FP=.5$). Session duration is not treated as time pressure, as the CoAuthor timer indicated a minimum, not a deadline.

3 Method

3.1 Quantitative model

For writer i and session j , $\bar{U}_i = n_i^{-1} \sum_j U_{ij}$ and $U_{ij}^W = U_{ij} - \bar{U}_i$ (likewise for H). We fit

$$\text{logit Pr}(O_{ij} > k) = \alpha_k + \beta_W U_{ij}^W + \beta_B \bar{U}_i + \gamma_W \frac{H_{ij}^W}{10} + \gamma_B \frac{\bar{H}_i}{10} + \delta_{p(ij)} + \theta_C R_{ij}^C + \theta_A R_{ij}^A, \quad k = 1, \dots, 6, \quad (1)$$

by stacking the cumulative binary indicators $\mathbb{I}(O_{ij} > k)$ and solving the working-independence ordinal GEE score (equivalently, a cumulative composite likelihood with common slopes). Uncertainty uses a writer-clustered sandwich covariance with $G/(G-1)$ correction and t_{G-1} intervals ($G=61$). $\exp(\beta_W)$ is the common cumulative odds ratio for a 1-point higher-than-usual U ; it is not a probability ratio or a causal effect. H2 replaces the outcome with I ; H3 adds $\bar{I}_i$, I_{ij}^W , and $U_{ij}^W I_{ij}^W$. The estimator was implemented directly in SciPy and checked numerically (analytic vs. finite-difference gradient 1.1×10^{-7} ; intercept-only model reproduces observed cumulative proportions to 8.5×10^{-10} ; FWL check of the linear FE coefficient to 3.6×10^{-16}).

Sensitivity analyses (added after initial results; reported in full, not selected for significance): equal total weight per writer ($w_{ij} \propto 1/n_i$); within-writer session order; usage and duration; additional post-hoc ratings (ideation, fluency); including zero-query sessions; earliest duplicate response; genre-specific fits; threshold-specific binary models $O > k$; and a linear writer fixed-effects model with a writer-level bootstrap (500 resamples, seed 20261009).

3.2 Qualitative contrast sampling

Algorithm 1 Contrastive case selection and preliminary coding

- 1: Cells $\leftarrow \{U \leq 3, U \geq 6\} \times \{O \leq 3, O \geq 6\} \times \{\text{creative, argumentative}\}$ $\triangleright$ 8 cells; ratings 4–5 excluded
 - 2: **for** each cell c **do**
 - 3: $S_c \leftarrow$ sessions in c with a non-empty improvement or free comment; shuffle with seed 20261009
 - 4: select up to 3 sessions, at most 1 per writer within c
 - 5: **end for**
 - 6: Read comments; assign multi-label codes (intent fit/mismatch, repetition, availability, ideation, choice, role)
 - 7: Record a paraphrase and an interpretive limit for each case; link to quantitative contrasts
 - 8: [**pending**] human re-coding, log replay of accept/revise/discard sequences
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This yielded 23 sessions from 16 writers. Coding was not blind (cases were selected on ratings), was produced by a single AI-assisted coder, and has not yet been reviewed by a human coder; we therefore report patterns, not prevalences, and do not claim a completed reflexive thematic analysis [6]. Comments mostly answer an “improvement” prompt, which biases toward criticism.

4 Results

4.1 Descriptives

Ownership was concentrated at the top of the scale (565 sessions at 7, 406 at 6). Means (SD) were similar across genres: ownership 5.87 (1.23) creative vs. 5.94 (1.16) argumentative; understanding 5.52 (1.52) vs. 5.39 (1.63); ideation help 5.47 (1.71) vs. 5.06 (1.89); human-written share 73.2% vs. 71.0%. Pooled, unadjusted Spearman correlations with ownership were small: understanding .058, ideation .105, human share .281.

4.2 Hypothesis tests

Table 2: Main conditional associations (cumulative odds ratios, writer-clustered 95% CI).

Outcome	Predictor	OR	95% CI	p
Ownership	U^W (H1)	0.847	[0.660, 1.086]	.187
Ownership	$\bar{U}$ (between)	2.356	[1.565, 3.547]	<.001
Ownership	H^W (+10 pp)	1.327	[1.130, 1.559]	<.001
Ideation help	U^W (H2)	3.396	[2.761, 4.176]	Holm <.001
Ownership	$U^W \times I^W$ (H3)	0.940	[0.876, 1.009]	Holm .087

H1 was not supported: the point estimate is below 1 and the interval includes 1. We do not reinterpret this as “understanding lowers ownership.” **H2 was supported** as an association between two post-hoc ratings; shared response style and semantic overlap of the items are plausible contributors. **H3 was not supported.** Within-writer human-written share was positively associated with ownership.

4.3 Robustness

Table 3: Within-writer understanding $\rightarrow$ ownership across specifications.

Specification	OR	95% CI	Specification	OR	95% CI
Primary	0.847	[0.660, 1.086]	+ fluency rating	0.832	[0.644, 1.074]
Equal writer weight	0.854	[0.735, 0.992]	Incl. zero-query	0.847	[0.659, 1.088]
+ session order	0.852	[0.663, 1.094]	Earliest duplicate	0.847	[0.660, 1.086]
+ usage, duration	0.839	[0.651, 1.082]	Creative only	0.927	[0.693, 1.240]
+ ideation rating	0.919	[0.768, 1.101]	Argumentative only	0.741	[0.572, 0.959]

No specification produced a positive within-writer association. Under equal-writer weighting the *between*-writer OR fell from 2.356 to 1.374 [0.978, 1.932], so the between-writer result depends on heavily participating writers and should not be read as a stable trait-level relationship. The linear writer-FE model gave -0.094 points $[-0.226, 0.037]$; bootstrap percentile CI $[-0.205, 0.028]$. Threshold-specific ORs declined from 1.016 ($O > 3$) to 0.741 ($O > 6$), suggesting that the proportional-odds summary may hide threshold heterogeneity; a formal test is pending. Negative estimates in single subgroups are not promoted to new confirmatory findings.

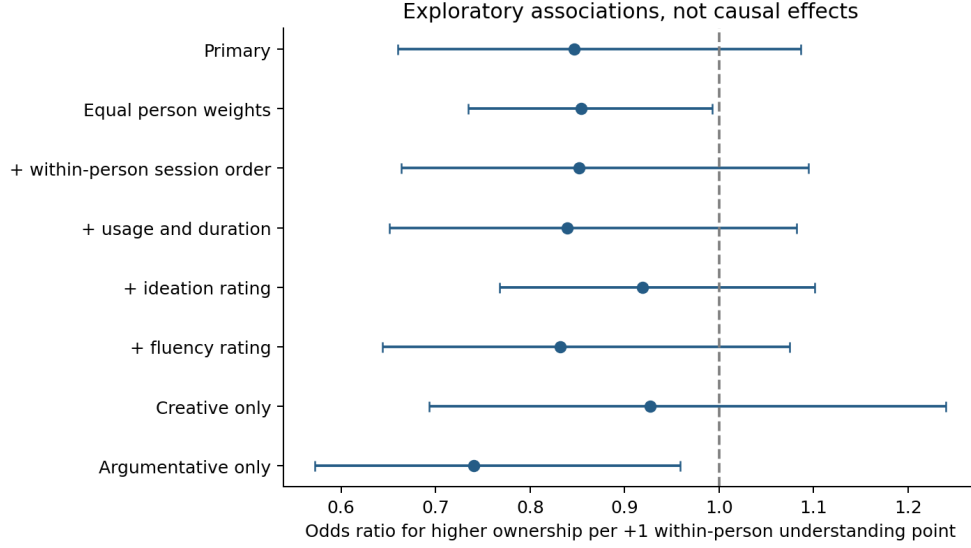


Figure 1: Within-writer understanding → ownership odds ratios across specifications (95% CI; dashed line = 1). From `reference_results/associations.png`.

4.4 Qualitative contrasts

Table 4: Illustrative contrastive cases (paraphrased, not quoted). $U/I/O$ on 1–7; H = human-written share.

Pattern	$U/I/O$	H	Writer’s account (paraphrase)	What it supports
Useful support, low ownership	7/7/2	39%	Used many suggestions; described self as closer to an editor than a writer.	Helpfulness, role, and ownership are distinct.
Mismatch, high ownership	3/2/7	89%	AI’s creative suggestions did not fit the intended direction.	Mismatch and ownership can co-occur.
Unstuck, low ownership	6/6/3	87%	Got stuck midway; AI’s new direction let them continue.	Help and low ownership coexist despite high human share.
Deliberate delegation	7/7/3	15%	Let the AI set topic and direction to test the system.	Low ownership need not mean unwanted loss of agency.
Questions scaffold expression	2/5/6	78%	Criticized latency but liked AI asking questions so they wrote sentences themselves.	A low U may reflect availability, not all support.
Choosing among options	6/5/6	66%	Could choose the story’s direction among several suggestions.	Candidate for measuring control directly.

Across cases, low understanding ratings often reflected service issues (no suggestion returned, latency) rather than semantic mismatch, and high helpfulness did not by itself imply ownership.

5 Discussion

Ideation help and ownership are different outcomes. Feeling understood tracked feeling helped, but not feeling ownership, within the same writer. Evaluations of writing assistants that collapse “positive experience” into one score would miss this divergence, just as individual creativity gains and collective diversity must be evaluated separately [5].

Role alignment as a follow-up question. The cases suggest that low ownership is not necessarily a bad outcome: a writer who chose to act as an editor may report low ownership accurately. A design goal may therefore be not to maximize ownership but to ensure that the *experienced role matches the desired role*. This hypothesis emerged from the analysis and has not been tested.

Unit of analysis matters. Session-weighted and writer-weighted estimates answer different questions (“a typical session” vs. “a typical writer”); with highly unequal participation, the choice changes the between-writer conclusion. Writer means are observational summaries, not personality measures.

Limitations. Single-item, same-time self-reports cannot identify causal order; human-written share may itself be a consequence of the collaboration, so the adjusted β_W is not a total effect. Ratings are ceiling-heavy and proportional odds may not hold. Sixty-one clusters limit between-writer and interaction estimates. The data reflect paid crowdworkers using GPT-3 in 2021, limiting generalization. Qualitative coding is preliminary and awaits human review; the 63/61 writer and 5/7-point scale discrepancies remain unresolved. No claims are made about authenticity, motivation, or mental health.

6 Future Work

1. **Process analysis of raw logs.** Reconstruct query $\rightarrow$ review $\rightarrow$ accept/discard $\rightarrow$ revise sequences; define time-to-first-query (with right-censoring) and pre-query pauses (10/20/30s sensitivity), excluding API-inserted text when measuring subsequent human writing. Pauses are not equated with “being stuck.”
2. **Descriptive state-transition model and microsimulation.** Estimate writer- and prompt-aware transition models, validate on held-out writers, and simulate trajectories only after the model reproduces observed behavior; simulated scenarios are not reported as intervention effects.
3. **Experiment.** A 2×2 design crossing an intent-confirmation interface (goal/perspective check vs. time-matched neutral preparation) with time budget (tight vs. generous), measuring ownership (primary), ideation help, perceived control, blind-rated creativity, and desired vs. experienced role; preregistered, intention-to-treat, with simulation-based power analysis rather than transplanting observational ORs.

Reproducibility. `run_pipeline.py` executes linkage, modeling, qualitative joining (hash-checked against source comments), and table generation; `reference_results/` contains the outputs reported here. Raw data are re-downloaded from the public source.

References

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